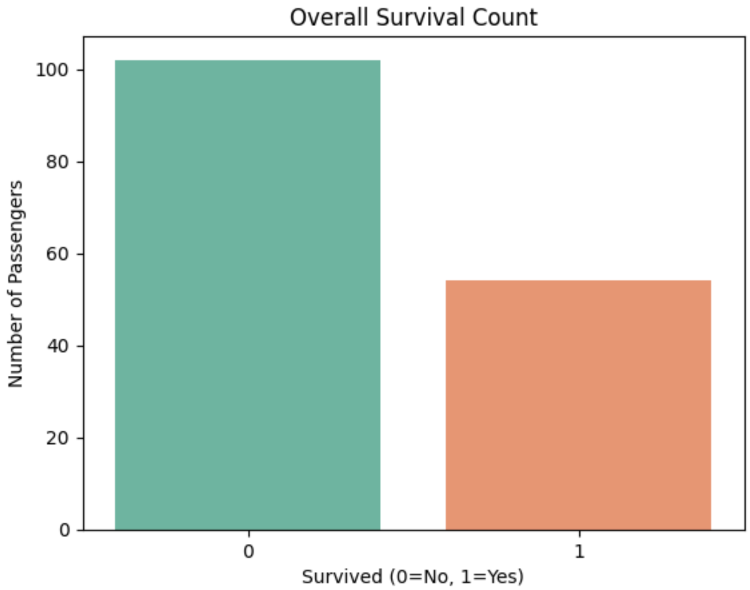
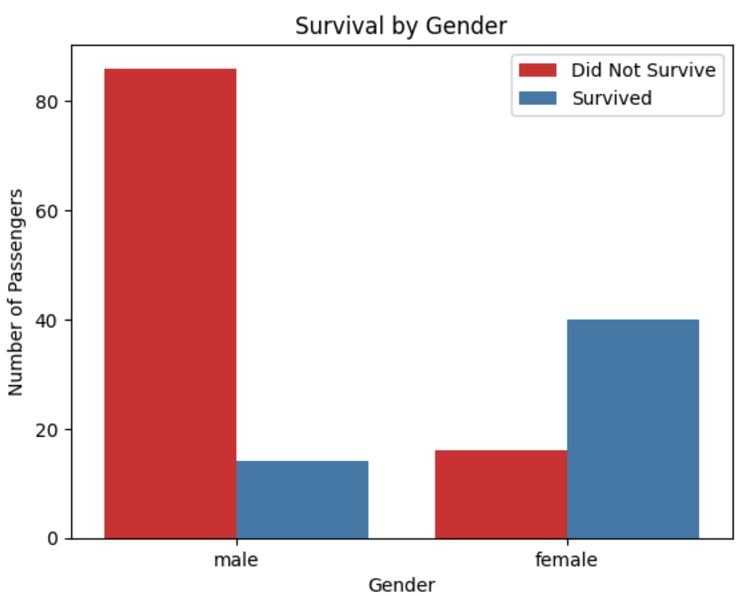
```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv("titanic.csv", delimiter="\t")
sns.countplot(x="Survived", data=df, palette="Set2")
plt.title("Overall Survival Count")
plt.xlabel("Survived (0=No, 1=Yes)")
plt.ylabel("Number of Passengers")
plt.show()
```

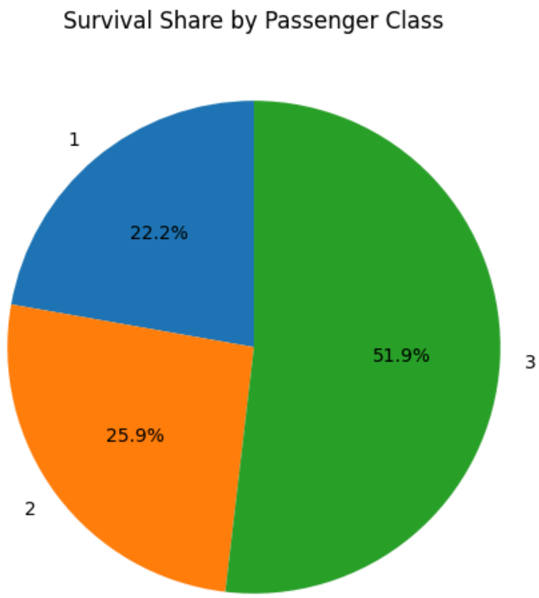
```
/tmp/ipykernel_751/762205022.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.countplot(x="Survived", data=df, palette="Set2")
```




```
sns.countplot(x="Sex", hue="Survived", data=df, palette="Set1")
plt.title("Survival by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.legend(["Did Not Survive", "Survived"])
plt.show()
```

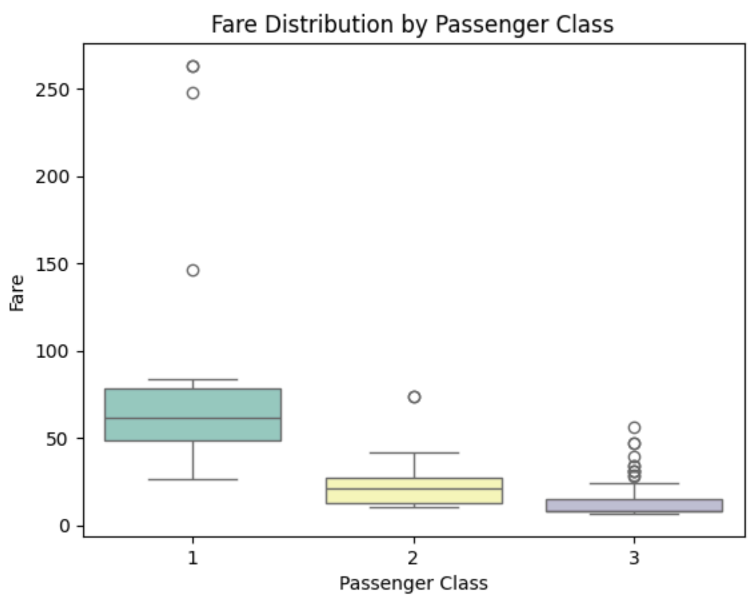


```
class_survival = df.groupby("Pclass")["Survived"].sum()
class_survival.plot(kind="pie", autopct="%1.1f%%", startangle=90, figsize=(6,6))
plt.title("Survival Share by Passenger Class")
plt.ylabel("")
plt.show()
```



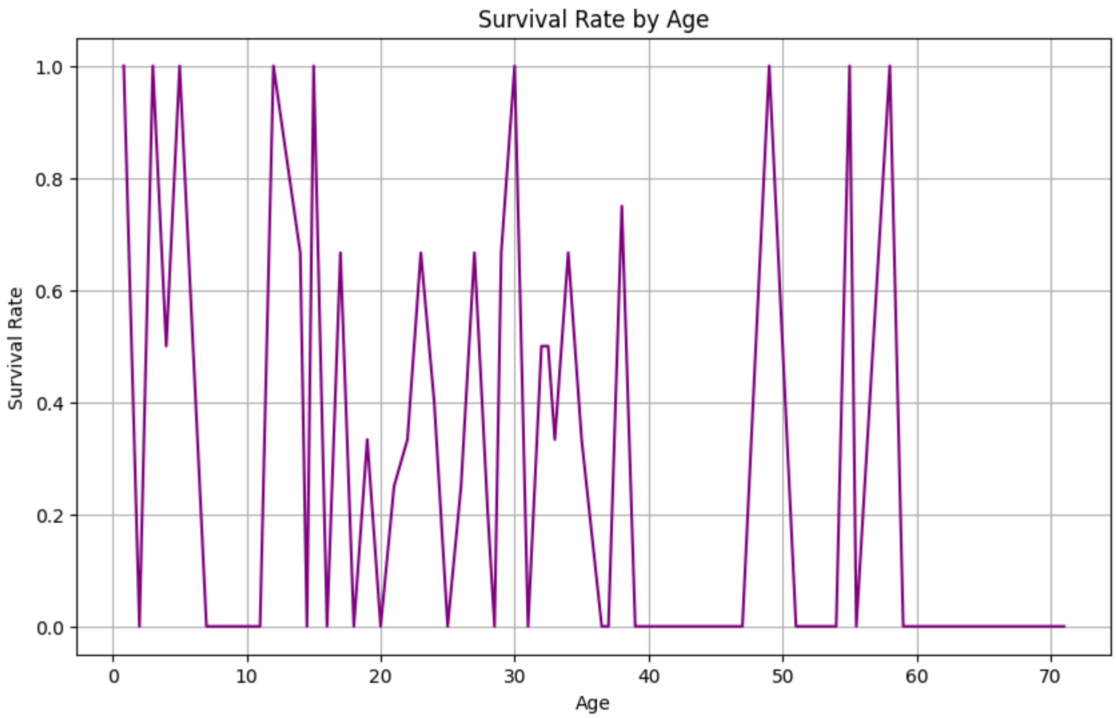

```
sns.boxplot(x="Pclass", y="Fare", data=df, palette="Set3")
plt.title("Fare Distribution by Passenger Class")
plt.xlabel("Passenger Class")
plt.ylabel("Fare")
plt.show()
```

```
/tmp/ipykernel_751/2834226623.py:1: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.boxplot(x="Pclass", y="Fare", data=df, palette="Set3")
```




```
age_survival = df.groupby("Age")["Survived"].mean()
age_survival.plot(kind="line", figsize=(10,6), color="purple")
plt.title("Survival Rate by Age")
plt.xlabel("Age")
plt.ylabel("Survival Rate")
```

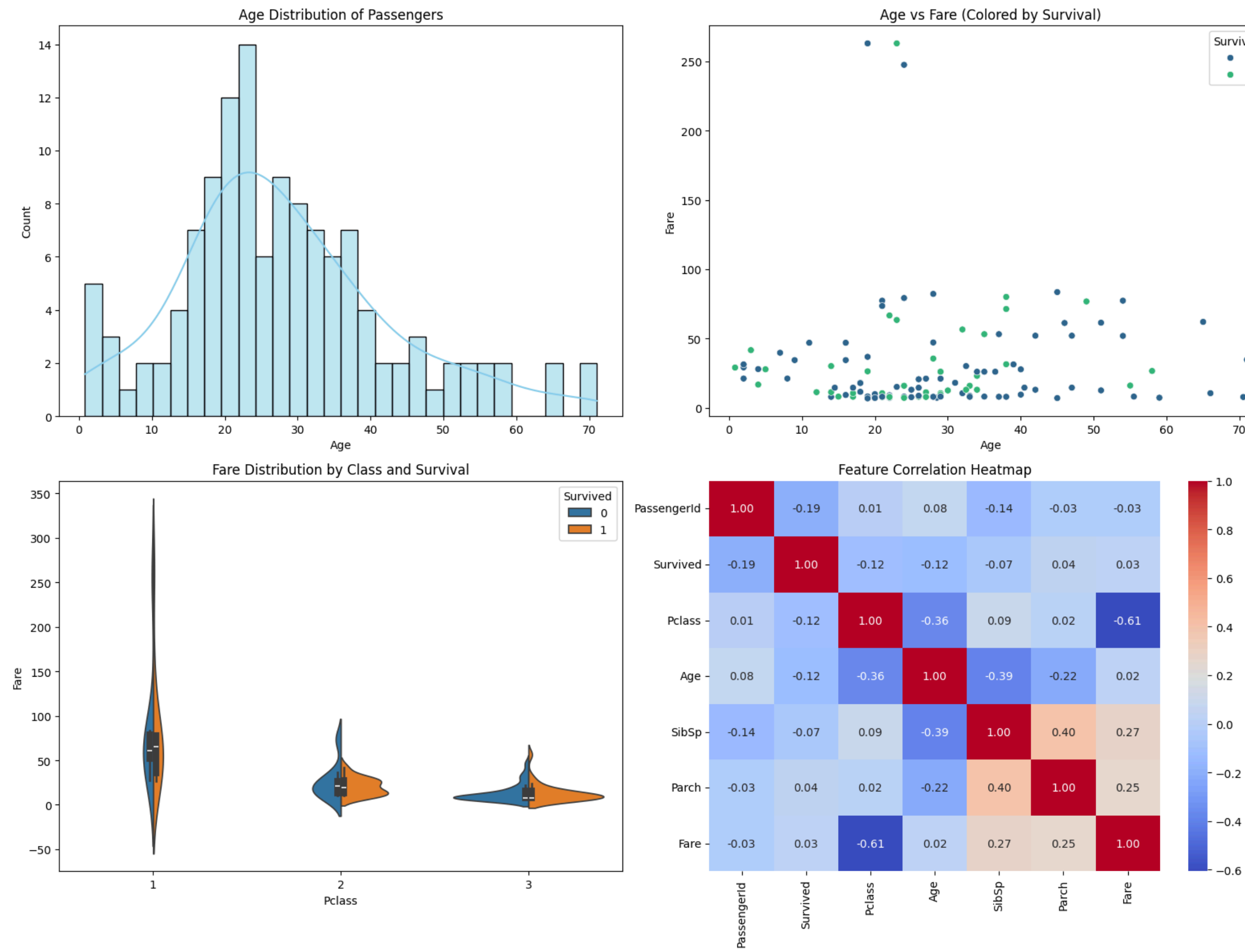
```
plt.grid(True)
plt.show()
```



Comprehensive Data Visualizations

Below are several different types of plots to explore the Titanic dataset from different perspectives.

```
import numpy as np

# Set up a large figure for multiple subplots
fig, axes = plt.subplots(2, 2, figsize=(16, 12))

# 1. Histogram: Age Distribution
sns.histplot(df['Age'].dropna(), bins=30, kde=True, ax=axes[0, 0], color='skyblue')
axes[0, 0].set_title('Age Distribution of Passengers')

# 2. Scatter Plot: Age vs Fare
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df, ax=axes[0, 1], palette='viridis')
axes[0, 1].set_title('Age vs Fare (Colored by Survival)')

# 3. Violin Plot: Fare distribution by Class and Survival
sns.violinplot(x='Pclass', y='Fare', hue='Survived', data=df, split=True, ax=axes[1, 0])
axes[1, 0].set_title('Fare Distribution by Class and Survival')

# 4. Heatmap: Correlation Matrix
corr = df.select_dtypes(include=[np.number]).corr()
sns.heatmap(corr, annot=True, cmap='coolwarm', fmt='.2f', ax=axes[1, 1])
axes[1, 1].set_title('Feature Correlation Heatmap')

plt.tight_layout()
plt.show()
```




```
# 5. Bar Plot: Survival Rate by Embarkation Point
plt.figure(figsize=(8, 5))
sns.barplot(x='Embarked', y='Survived', data=df, palette='pastel', ci=None)
plt.title('Survival Rate by Port of Embarkation')
plt.ylabel('Survival Probability')
plt.show()
```

```
/tmp/ipykernel_751/1002144258.py:3: FutureWarning:

The `ci` parameter is deprecated. Use `errorbar=None` for the same effect.

  sns.barplot(x='Embarked', y='Survived', data=df, palette='pastel', ci=None)
/tmp/ipykernel_751/1002144258.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v0.14.0. Assign the `x` variable to `hue` and set

  sns.barplot(x='Embarked', y='Survived', data=df, palette='pastel', ci=None)
```